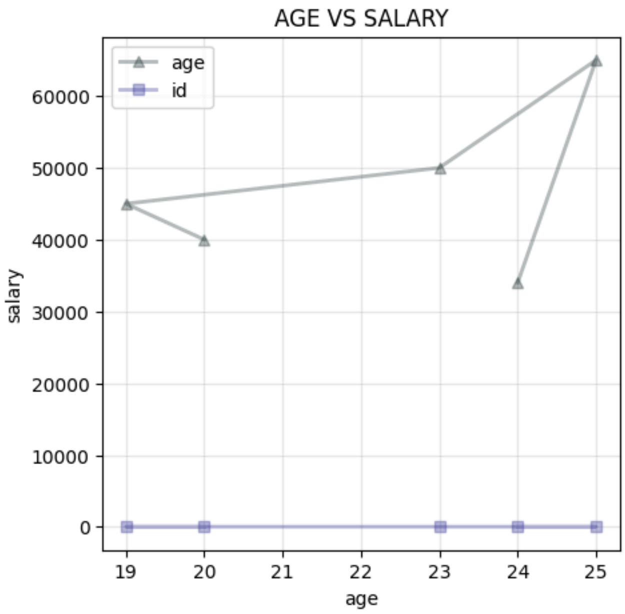
```
!pip install matplotlib seaborn pandas numpy
```

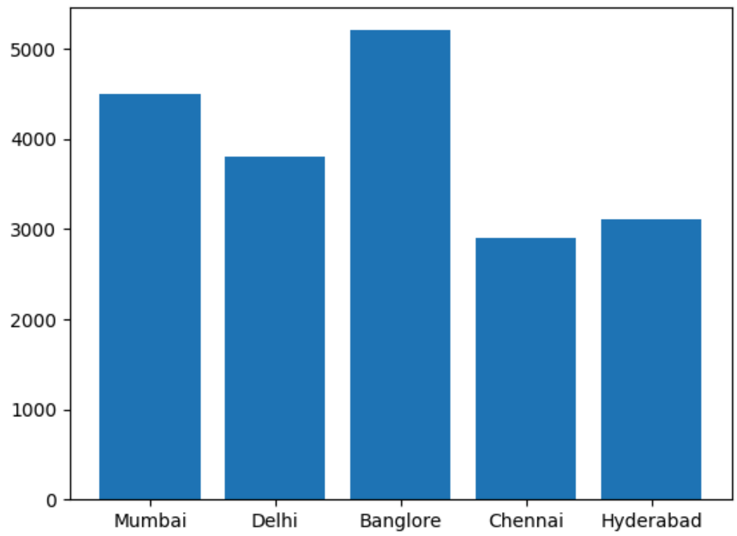
```
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-packages (0.13.2)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.62.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.5.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (26.1)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (2.9.0.post
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2026.1)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib) (
```

```
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
```

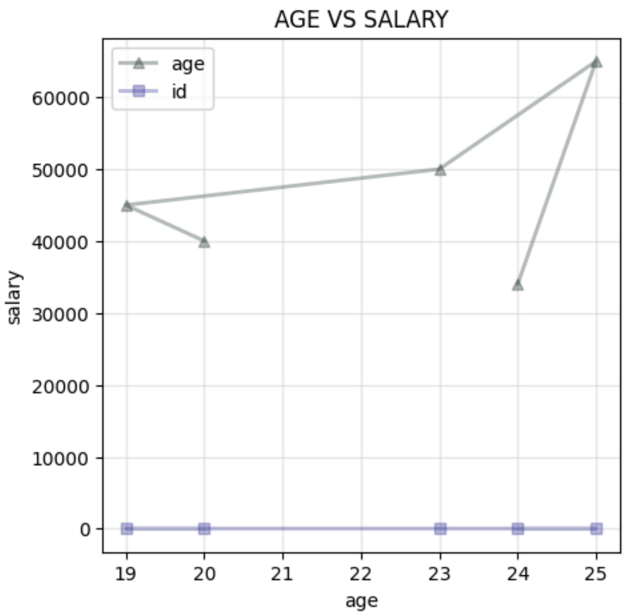
```
age=[20,19,23,25,24]
salary=[40000,45000,50000,65000,34000]
id=[30,31,29,28,32]
plt.figure(figsize=(5,5))
plt.plot(age,salary,marker="^",label="age",linewidth=2,color="#2335")
plt.plot(age,id,marker='s',label="id",linewidth=2,color="#3395")
plt.legend()
plt.xlabel("age")
plt.ylabel("salary")
plt.title("AGE VS SALARY")
plt.grid(True,alpha=0.3)
plt.savefig("age_vs_salary.png")
plt.show()
```




```
#bar chart
cities = ['Mumbai','Delhi','Banglore','Chennai','Hyderabad']
revenue = [4500, 3800, 5200, 2900, 3100]
fix, ax = plt.subplots()
ax.bar(cities,revenue)
plt.show()
```

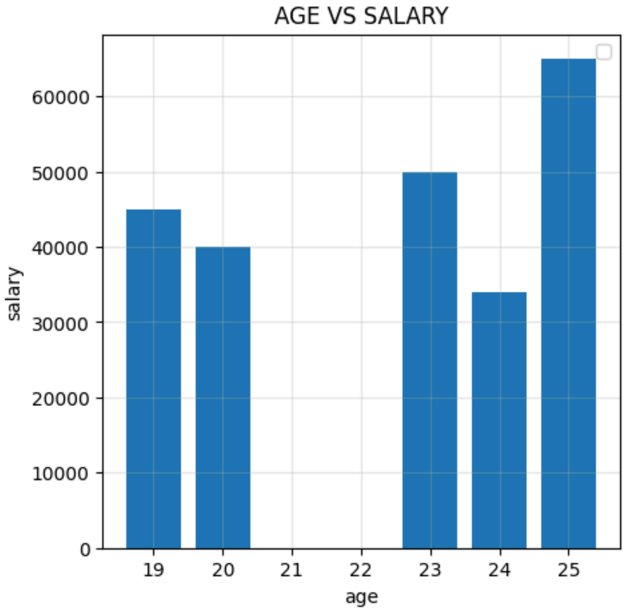



```
#Line Chart
age=[20,19,23,25,24]
salary=[40000,45000,50000,65000,34000]
plt.figure(figsize=(5,5))
plt.plot(age,salary,marker="^",label="age",linewidth=2,color="#2335")
plt.plot(age,id,marker='s',label="id",linewidth=2,color="#3395")
plt.legend()
plt.xlabel("age")
plt.ylabel("salary")
plt.title("AGE VS SALARY")
plt.grid(True,alpha=0.3)
plt.savefig("age_vs_salary.png")
plt.show()
```




```
#Bar Graph
age=[20,19,23,25,24]
salary=[40000,45000,50000,65000,34000]
plt.figure(figsize=(5,5))
plt.bar(age,salary)
plt.legend()
plt.xlabel("age")
plt.ylabel("salary")
plt.title("AGE VS SALARY")
plt.grid(True,alpha=0.3)
plt.savefig("age_vs_salary.png")
plt.show()
```

```
/tmp/ipykernel_1999/170659422.py:6: UserWarning: No artists with labels found to put in legend.  Note that artists whose lab
  plt.legend()
```



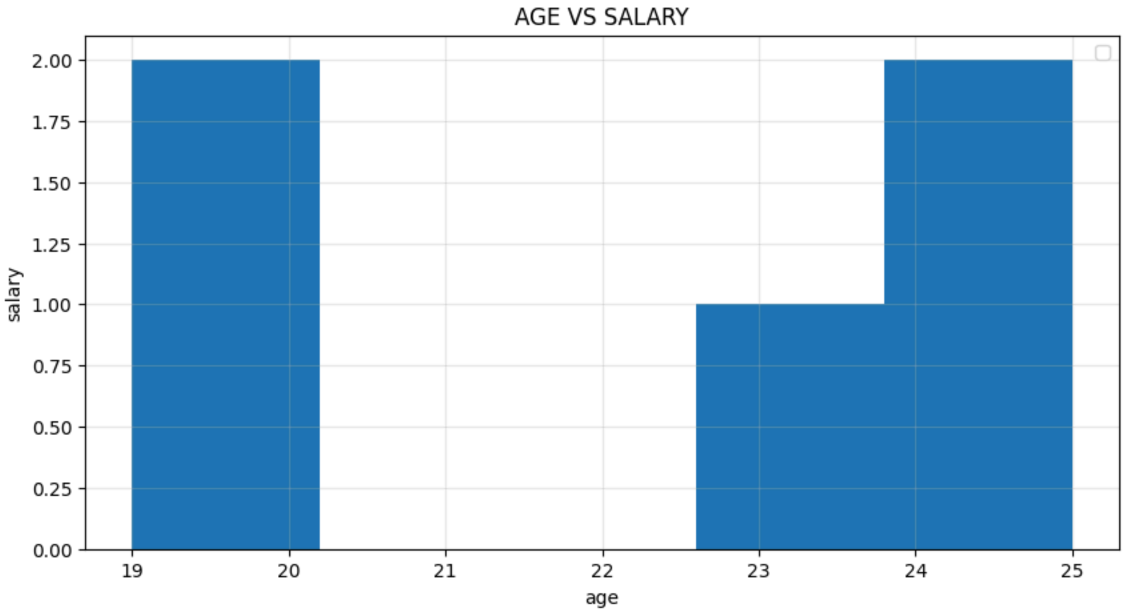

```
#Scatter
age=[20,19,23,25,24]
salary=[40000,45000,50000,65000,34000]
plt.figure(figsize=(5,5))
plt.scatter(salary,age)
plt.legend()
plt.xlabel("age")
plt.ylabel("salary")
plt.title("AGE VS SALARY")
plt.grid(True,alpha=0.3)
plt.savefig("age_vs_salary.png")
plt.show()
```

```
/tmp/ipykernel_1999/248068771.py:6: UserWarning: No artists with labels found to put in legend.  Note that artists whose lab
  plt.legend()
```


```
#histogram
age=[20,19,23,25,24]
salary=[40000,45000,50000,65000,34000]
plt.figure(figsize=(10,5))
plt.hist(age,bins=5)
plt.legend()
plt.xlabel("age")
plt.ylabel("salary")
plt.title("AGE VS SALARY")
plt.grid(True,alpha=0.3)
```

```
plt.savefig("age_vs_salary.png")
```

```
/tmp/ipykernel_1999/1916225529.py:6: UserWarning: No artists with labels found to put in legend.  Note that artists whose la
  plt.legend()
```




```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd
```

```
sns.set_theme(style="darkgrid")
df=sns.load_dataset('titanic')
print(df)
```

```
     survived  pclass     sex   age  sibsp  parch     fare embarked   class  \
0           0       3    male  22.0      1      0   7.2500        S   Third
1           1       1  female  38.0      1      0  71.2833        C   First
2           1       3  female  26.0      0      0   7.9250        S   Third
3           1       1  female  35.0      1      0  53.1000        S   First
4           0       3    male  35.0      0      0   8.0500        S   Third
..        ...     ...     ...   ...    ...    ...      ...      ...     ...
886         0       2    male  27.0      0      0  13.0000        S  Second
887         1       1  female  19.0      0      0  30.0000        S   First
888         0       3  female   NaN      1      2  23.4500        S   Third
889         1       1    male  26.0      0      0  30.0000        C   First
890         0       3    male  32.0      0      0   7.7500        Q   Third

       who  adult_male deck  embark_town alive  alone
0      man        True  NaN  Southampton    no  False
1    woman       False    C    Cherbourg   yes  False
2    woman       False  NaN  Southampton   yes   True
3    woman       False    C  Southampton   yes  False
4      man        True  NaN  Southampton    no   True
..     ...         ...  ...          ...   ...    ...
886    man        True  NaN  Southampton    no   True
887  woman       False    B  Southampton   yes   True
888  woman       False  NaN  Southampton    no  False
889    man        True    C    Cherbourg   yes   True
890    man        True  NaN   Queenstown    no   True

[891 rows x 15 columns]
```